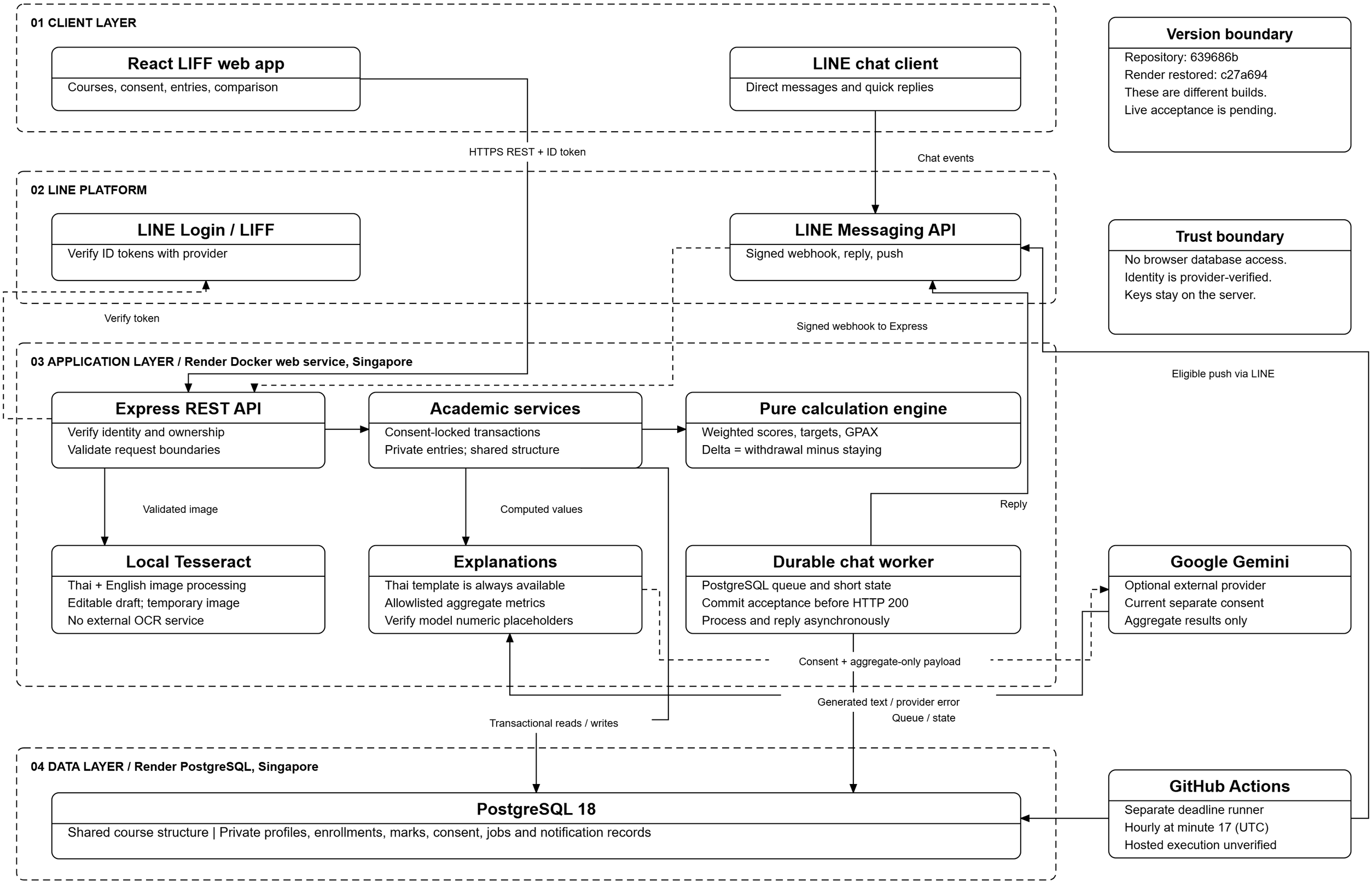
System architecture

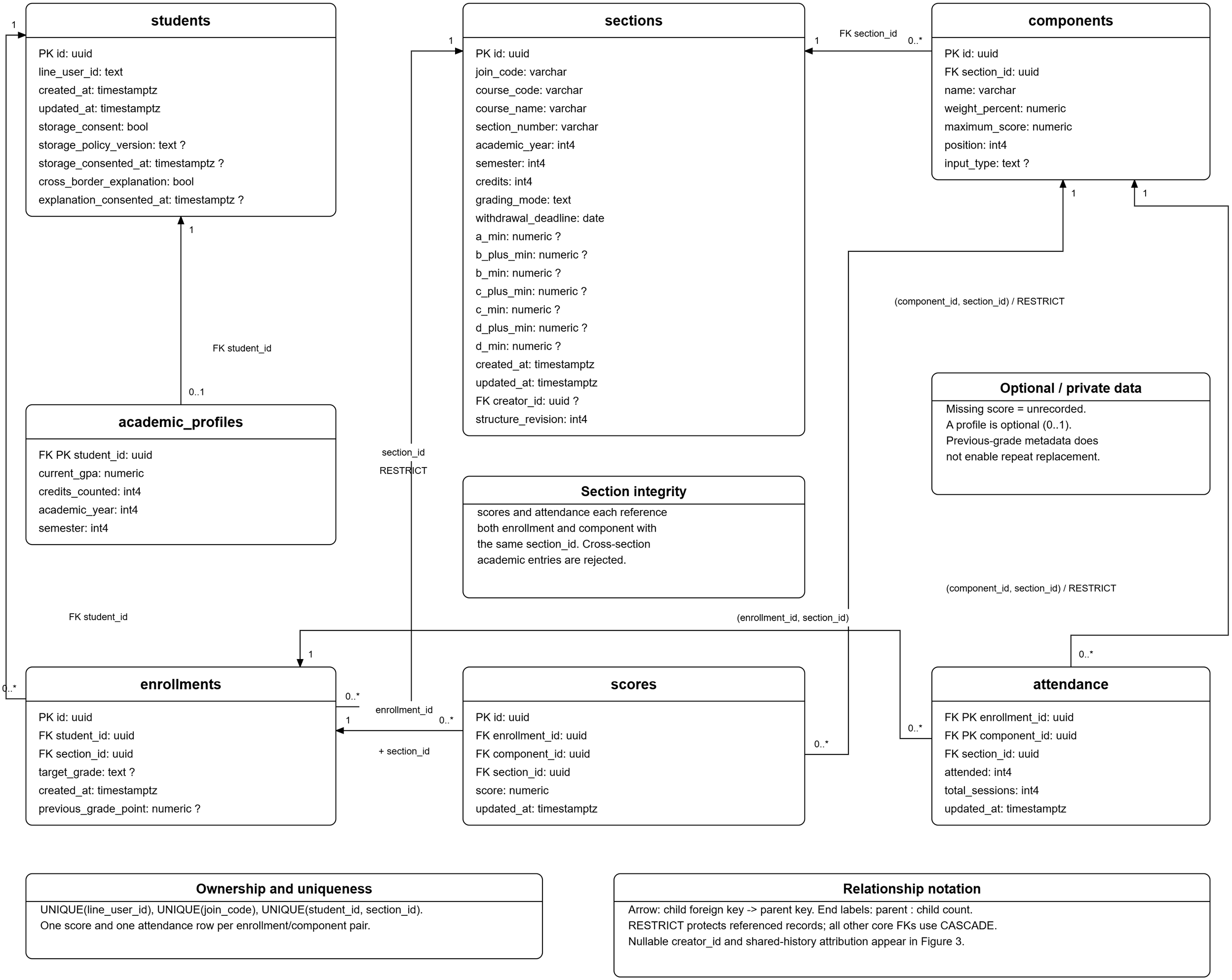
Four logical layers, physical hosting, and external processing boundaries.



Client | Application | Data | Identity / consent | Messages | Hosting / scheduler

Core entity-relationship diagram

Physical keys and ownership for shared course rules and private academic records.



Ownership and uniqueness

UNIQUE(line_user_id), UNIQUE(join_code), UNIQUE(student_id, section_id).
One score and one attendance row per enrollment/component pair.

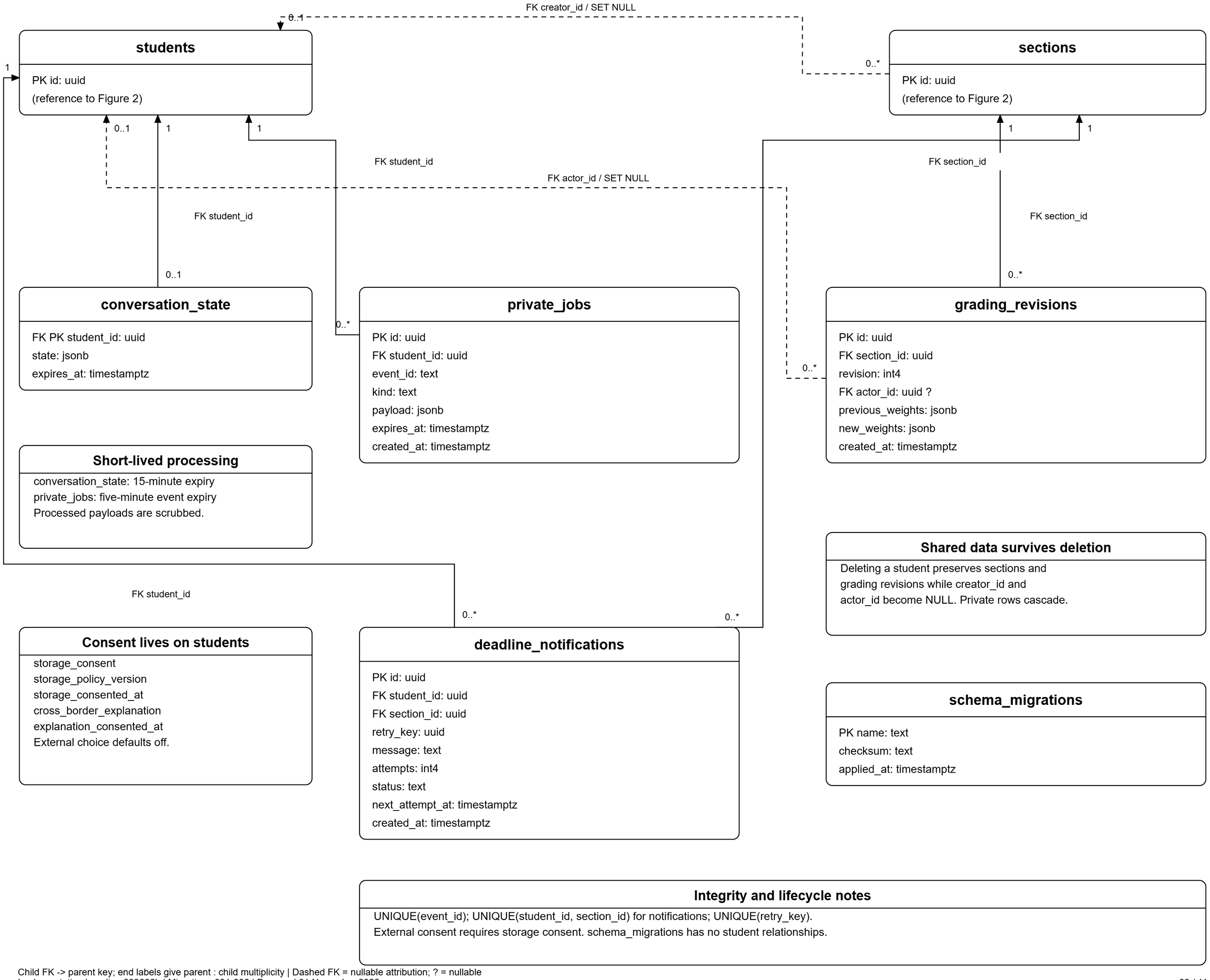
Relationship notation

Arrow: child foreign key -> parent key. End labels: parent : child count.
RESTRICT protects referenced records; all other core FKs use CASCADE.
Nullable creator_id and shared-history attribution appear in Figure 3.

PK = primary key; FK = foreign key; ? = nullable | Application additionally enforces consent and ownership

Supporting entity-relationship diagram

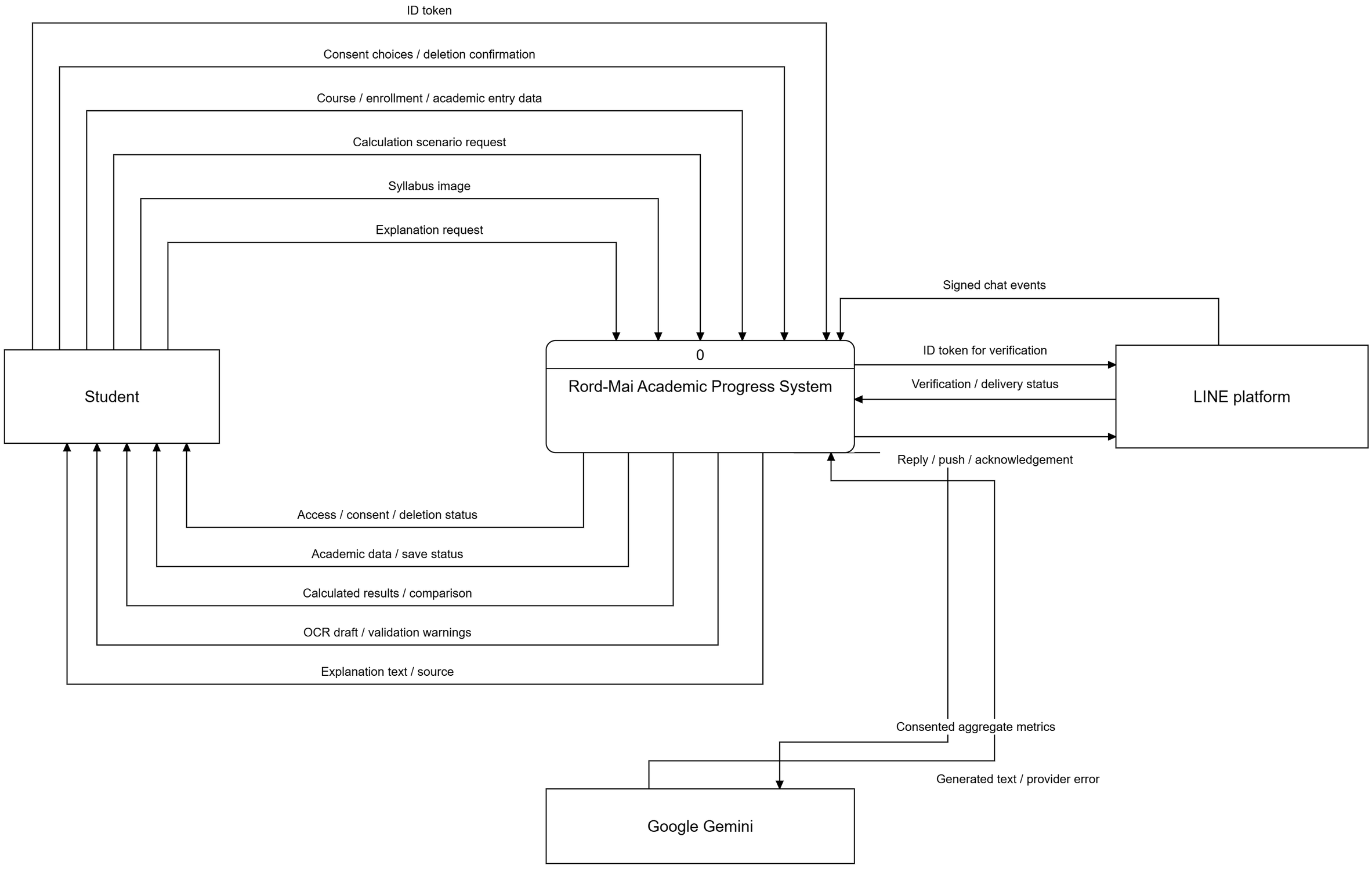
Consent evidence, deletion behaviour, asynchronous processing and schema bookkeeping.



Child FK -> parent key; end labels give parent : child multiplicity | Dashed FK = nullable attribution; ? = nullable
Implementation baseline 639686b | Migrations 001-006 | Prepared 01 November 2026

Level-0 DFD / Context Diagram of Rord-Mai

One system process and the data crossing its boundary. No internal data stores.

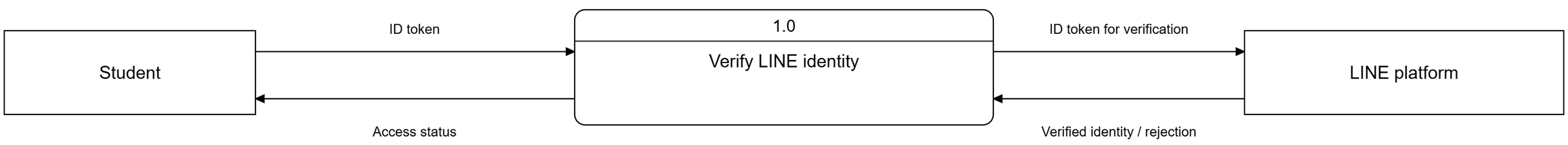


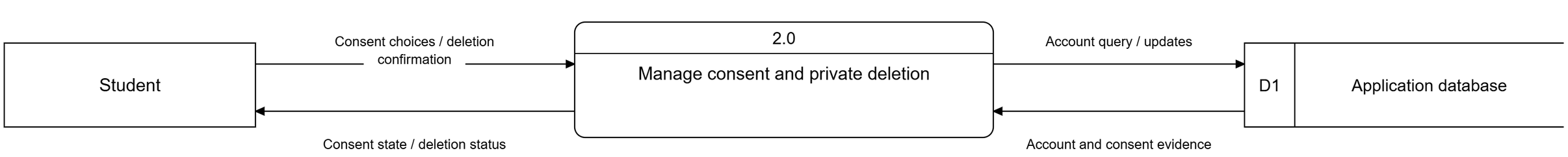
Notation: numbered rounded process | square external entity | open-ended data store | labelled data flow

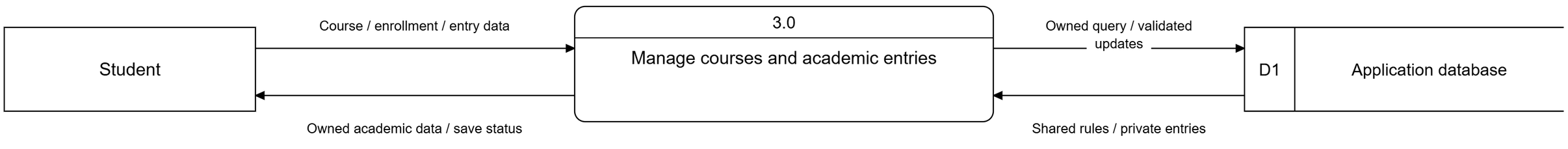
Repeated entity names and D1 symbols refer to the same external entities or application data store.

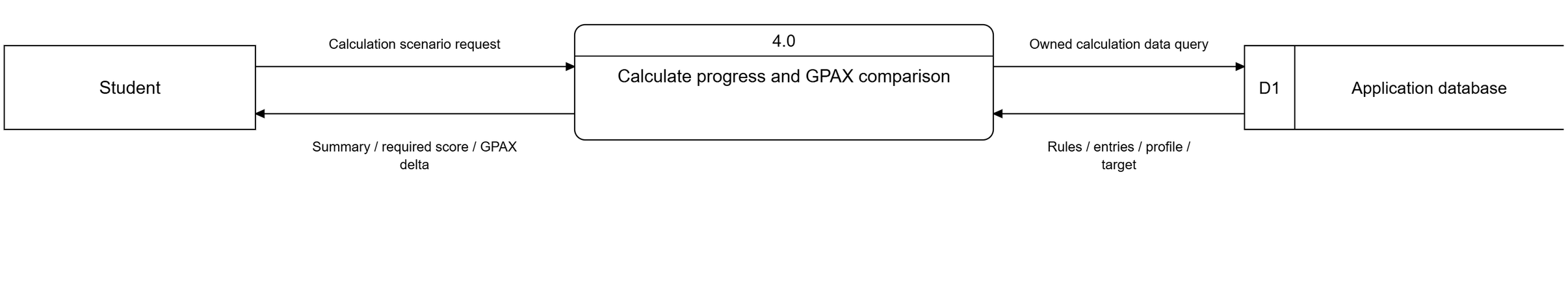
Level-1 DFD of Rord-Mai

Eight main processes. D1 is one logical application data store backed by PostgreSQL.

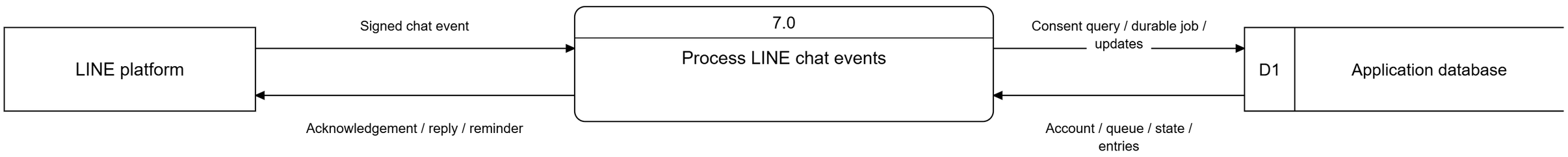




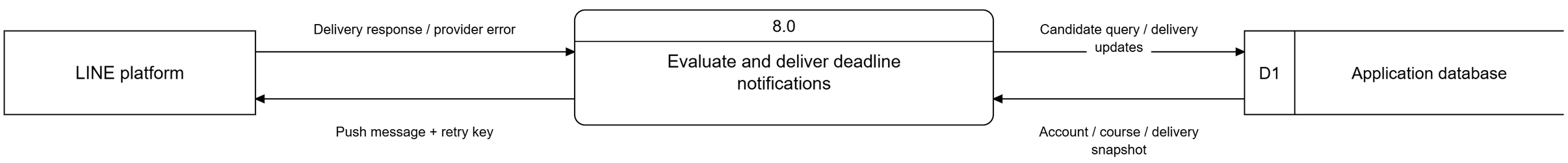




Process 6.0 sends only aggregate metrics externally; declined or invalid output uses the local template.



Process 7.0 stores consented work only. A consentless reminder stays request-local and follows acknowledgement.



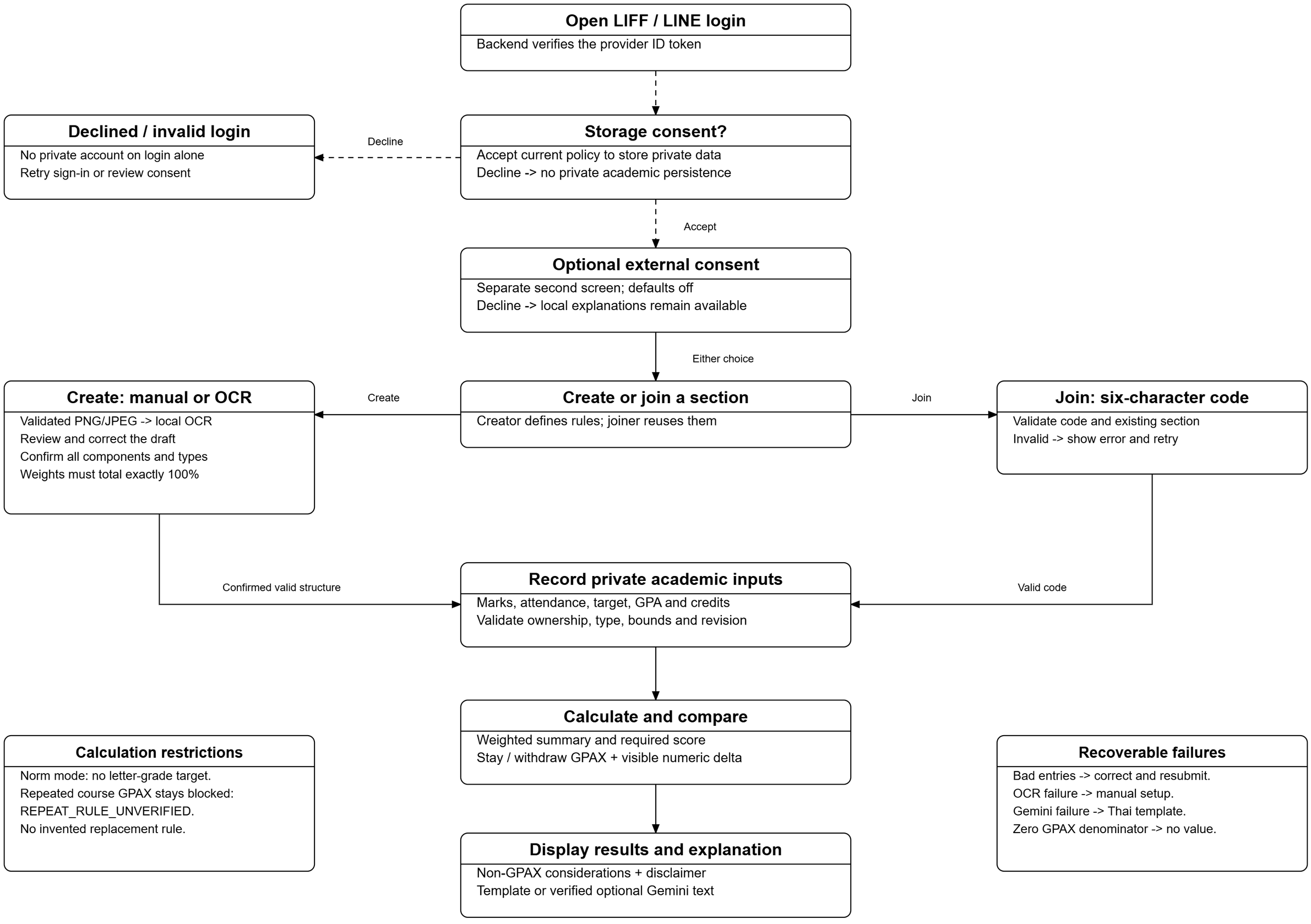
D1 maps to students, sections, components, enrollments, scores, attendance, academic_profiles, grading_revisions, conversation_state, private_jobs and deadline_notifications. Technical schema_migrations is omitted from domain flow.

Notation: numbered rounded process | square external entity | open-ended data store | labelled data flow

Repeated entity names and D1 symbols refer to the same external entities or application data store.

System workflow

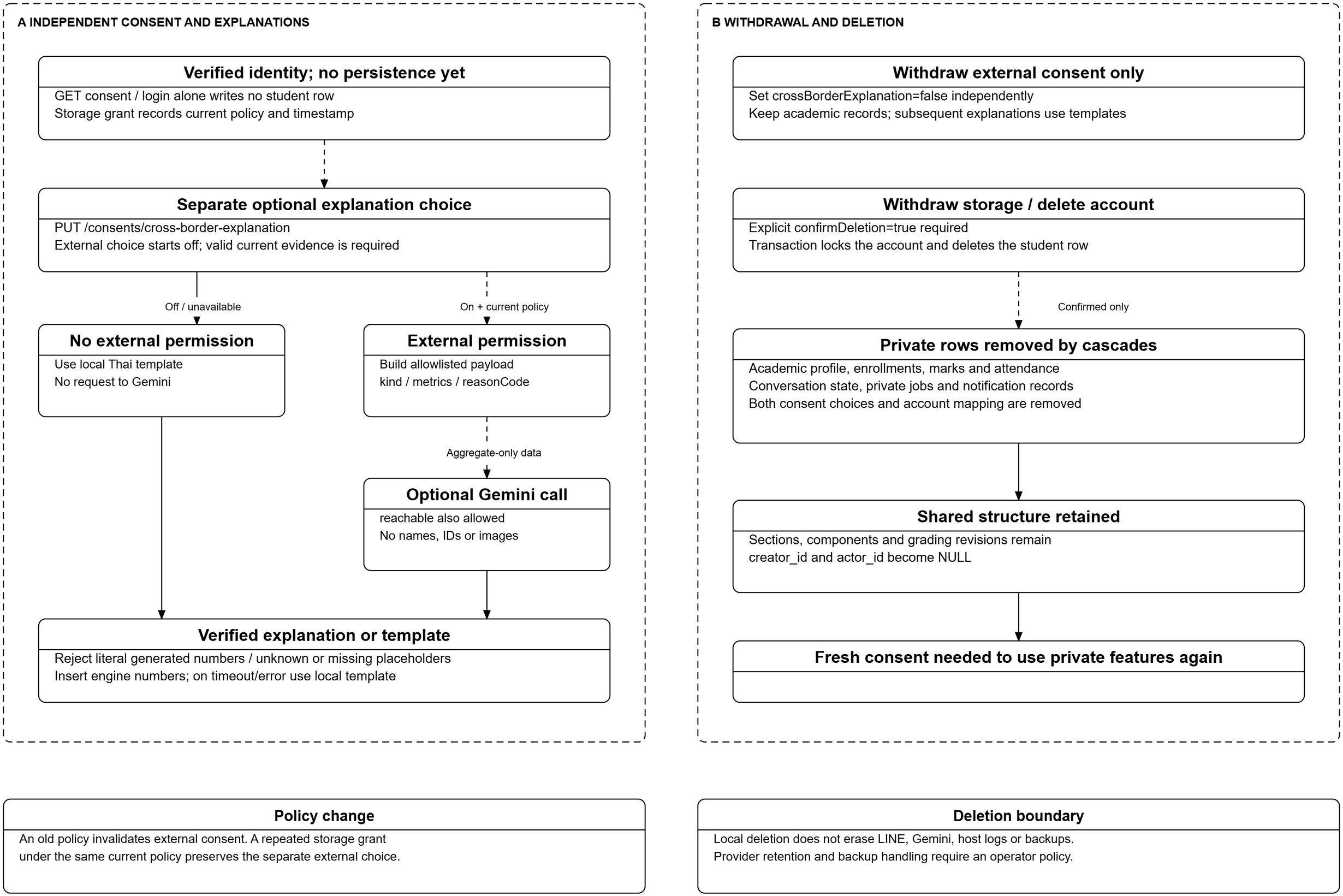
Main student journey, with validation loops and safe alternatives.



Notifications are a separate scheduled flow, not a required step of this screen journey (Figure 8).

Consent and data lifecycle

Independent choices, bounded external processing, and confirmed private deletion.



Policy change

An old policy invalidates external consent. A repeated storage grant under the same current policy preserves the separate external choice.

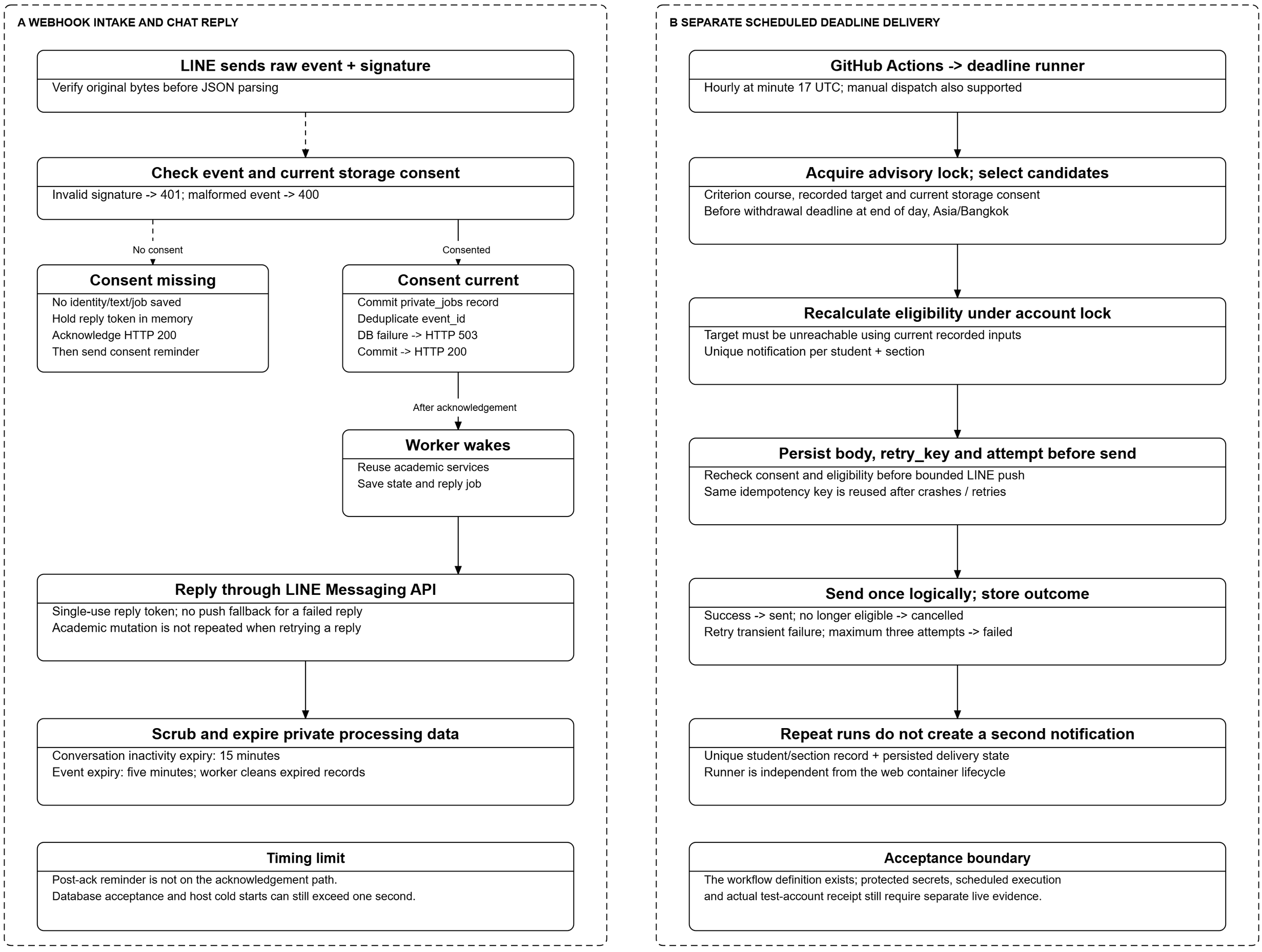
Deletion boundary

Local deletion does not erase LINE, Gemini, host logs or backups. Provider retention and backup handling require an operator policy.

Consent / confirmation | Database lifecycle | Local processing | External provider

LINE intake and deadline delivery

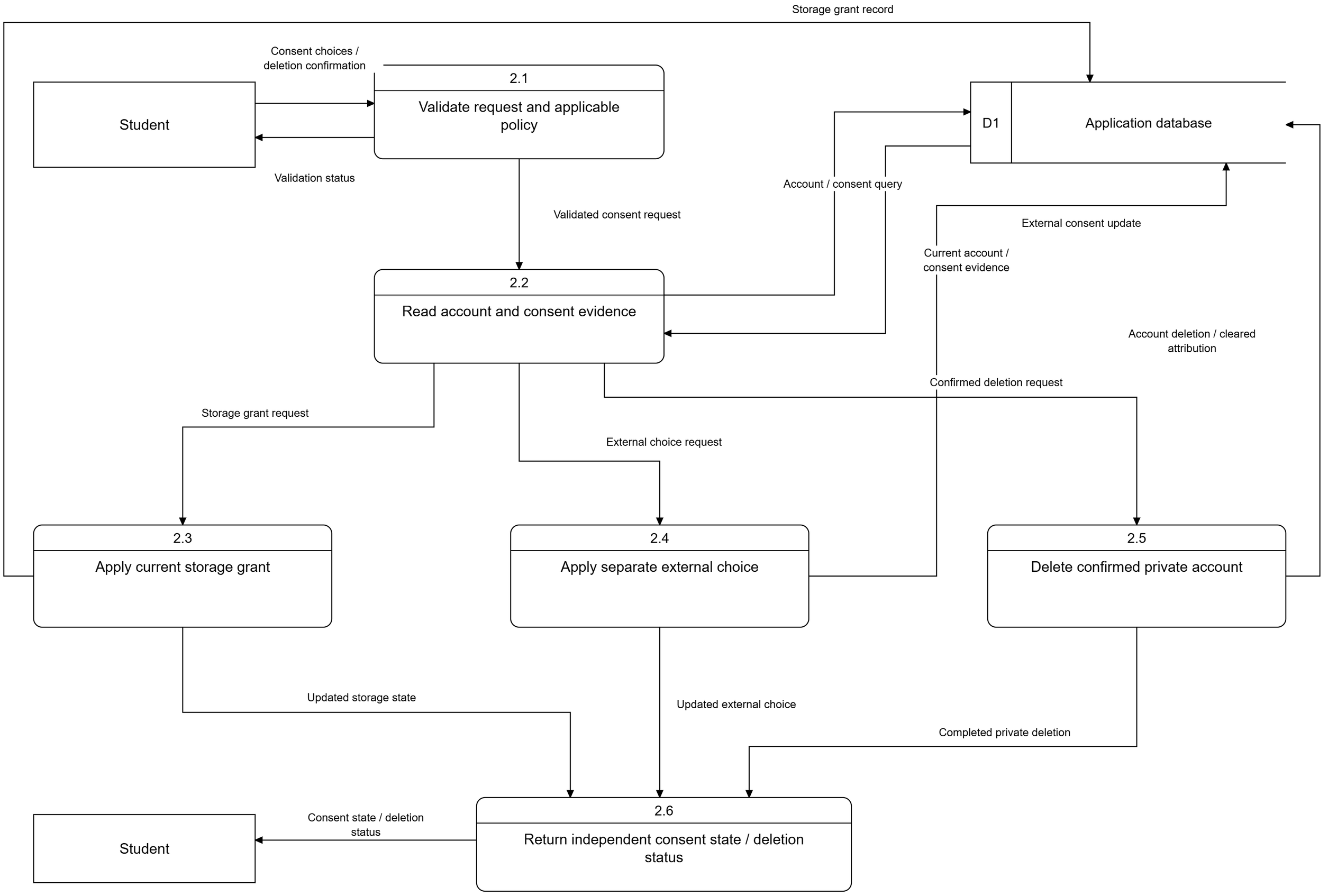
Durable chat acceptance and a separate, idempotent scheduled notification runner.



LINE messages / events | Durable state | Verification / consent | External scheduler

Level-2 DFD for Process 2.0: Manage Consent

Parent process 2.0 is expanded into 2.1-2.6 without changing its external data contract.



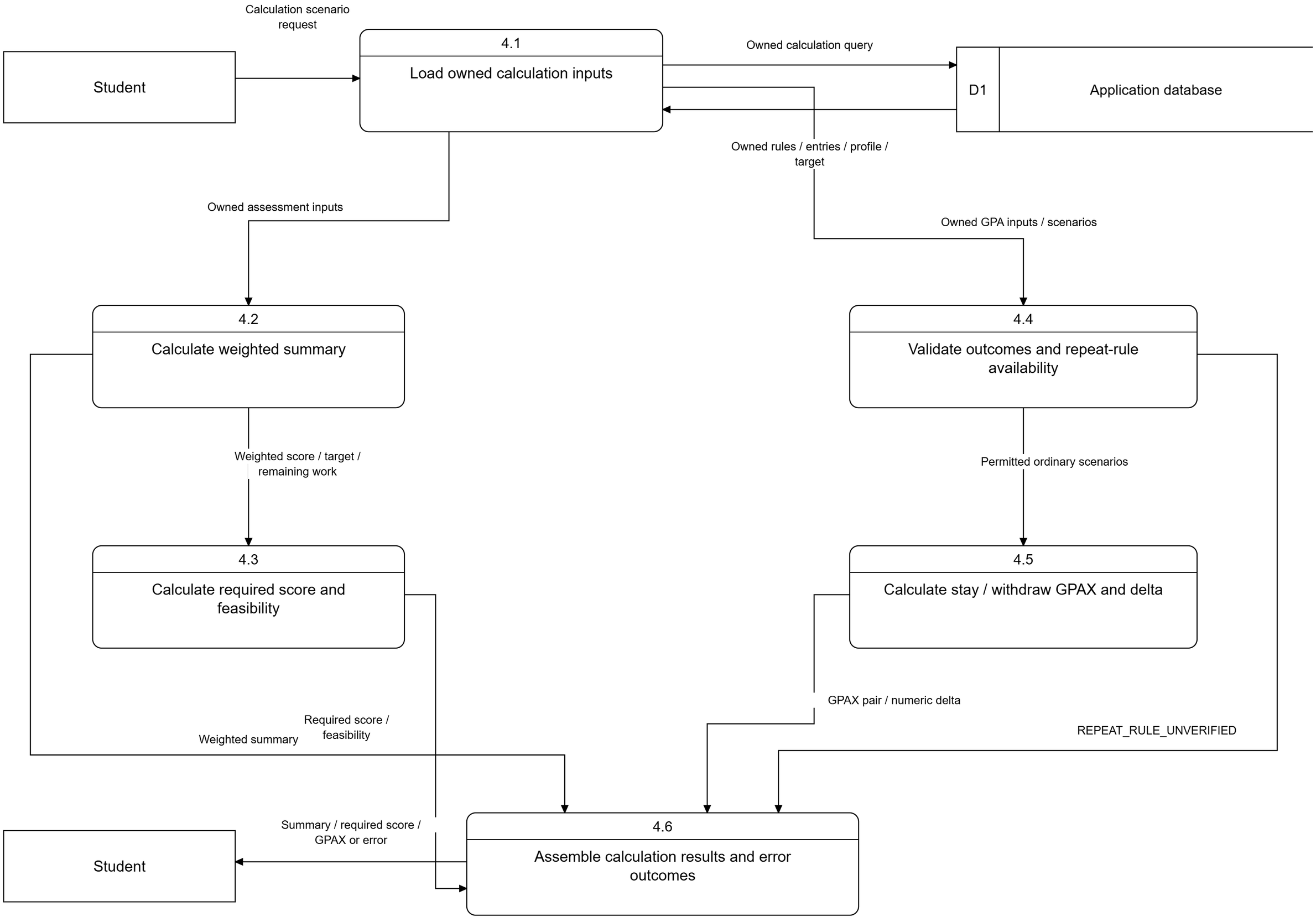
Current-policy checks apply to grants/updates. Deletion requires explicit confirmation. External withdrawal preserves academic data.

Notation: numbered rounded process | square external entity | open-ended data store | labelled data flow

Repeated entity names and D1 symbols refer to the same external entities or application data store.

Level-2 DFD for Process 4.0: Calculate Results

Parent process 4.0 is expanded into 4.1-4.6; pure arithmetic remains inside the calculation service.



The service loads data; the pure engine has no database I/O. Delta = withdrawal GPAX minus staying GPAX.

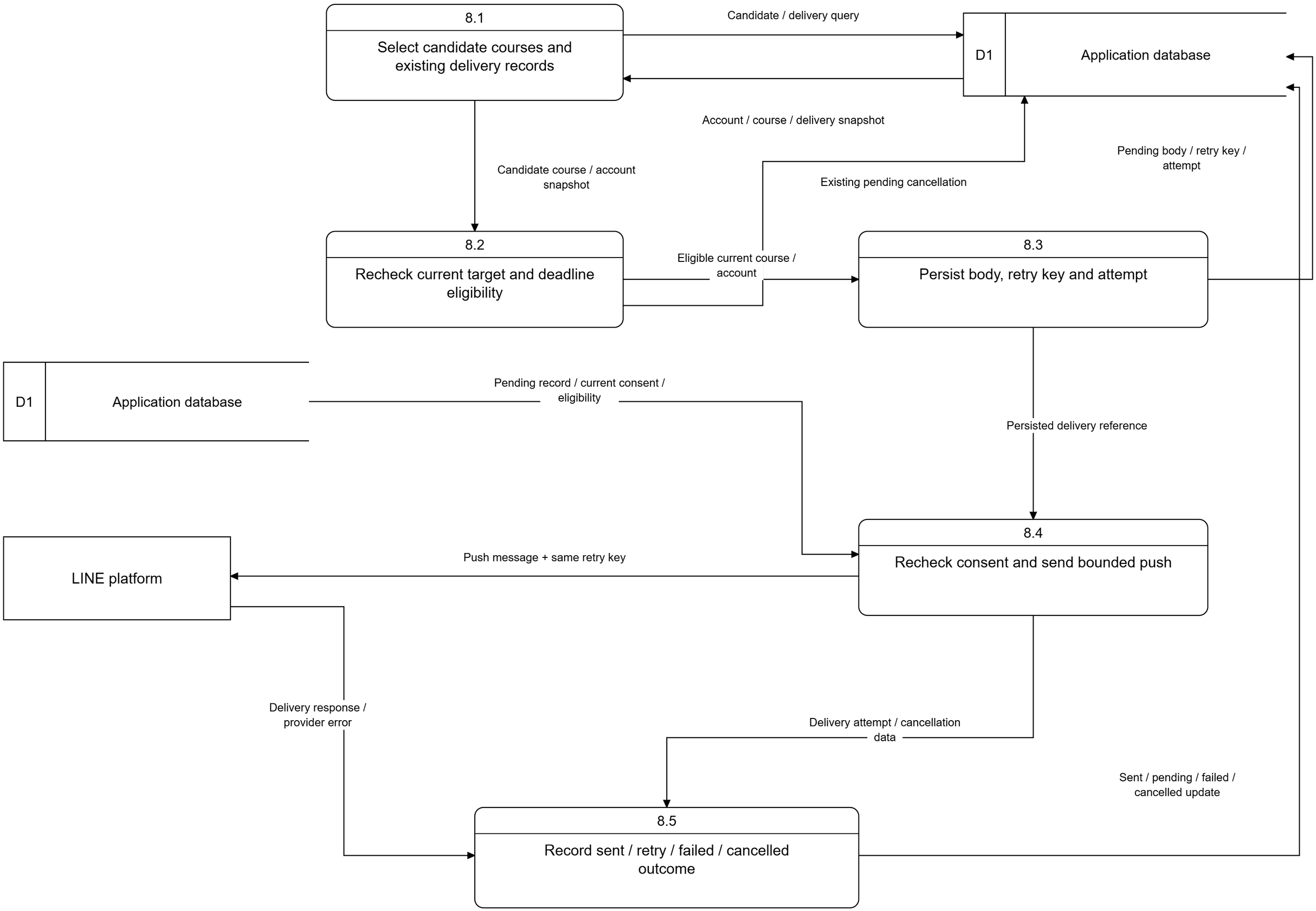
Norm-referenced targets and zero-denominator GPAX are returned as unavailable/error outcomes, not invented numeric results.

Notation: numbered rounded process | square external entity | open-ended data store | labelled data flow

Repeated entity names and D1 symbols refer to the same external entities or application data store.

Level-2 DFD for Process 8.0: Deadline Notifications

Parent process 8.0 is expanded into 8.1-8.5; a schedule triggers execution, not a DFD control arrow.



D1 persists retry keys and unique student/section records. Retriable failures are bounded to three attempts.

Notation: numbered rounded process | square external entity | open-ended data store | labelled data flow

Repeated entity names and D1 symbols refer to the same external entities or application data store.